Design and Fabrication of an RF Gun for the Canadian Light Source

1.8.32.3 – Rev 0

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TABLE OF CONTENTS

1.0 PURPOSE

This agreement describes the specifications for the thermionic RF gun to be delivered by MAX IV Laboratory (MAX IV Laboratory) to the Canadian Light Source. (CLS).

2.0 SCOPE

The RF gun is to be supplied under a sole source agreement with MAX IV Laboratory as part of a collaborative effort defined in a Memorandum of Understanding between MAX IV Laboratory and the CLS. As such the scope of this agreement is limited to the technical specification and other considerations associated with the usual tendering process are waived. Delivery and financial terms are for in the general contract information.

The contract to build an RF gun shall include a design modification to be done by MAX IV Laboratory and approved by CLS. As such the specifications that follow are only representative of the expected performance based on the 3000 MHz RF gun in use at MAX IV Laboratory. Final specifications will be approved after the design phase is complete.

Once CLS has approved the MAX IV Laboratory design specifications for a 2856 MHz RF gun MAX IV Laboratory will construct a new gun to meet the final specifications.

3.0 GENERAL BACKGROUND

The present DC thermionic gun for the CLS linac is close to 30 years old. Although it continues to reliably produce the electron pulses for the required linac operations, there is a concern that aging components may fail and be difficult to repair or replace. At the same time a new linac configuration is under study that would use four new 2856 MHz RF sections that are presently on site. With the addition of a new gun a virtually new linac could be installed in the near future. The low energy of the present gun requires that the present linac has to have a graded-β section to bring electrons to relativistic energies. There is no new section to replace this aging section. The advantage of an RF gun is that the output energy is sufficiently relativistic that graded-β section is no longer required.

MAX IV Laboratory (formerly MAX-Laboratory) has built and operated several thermionic RF gun over the last 10 years or so. An RF gun has reliably provided electrons to the MAX-Laboratory storage rings for many years. As well, RF guns are now under development for the new MAX IV linac. The RF guns for MAX IV Laboratory operate at 3000 MHz. MAX IV Laboratory personnel have indicated that they have the expertise to modify their gun design to 2856 MHz operation suitable for use with the CLS linac. As well, they have the appropriate facilities available to produce the gun in a timely manner.

4.0 DEFINITIONS

Shall: Indicates a requirement that is to be strictly followed in order to conform to the specification.

Should: Indicates a recommended or preferred requirement. Deviation must be justified.

May: Indicates a possible method or course of action.

5.0 REQUIREMENTS

A schematic cross-section of the run gun is shown in figure 1. As seen in the figure the main features of the gun are the resonant RF cavity, the RF-choke and the thermionic cathode. The thermionic cathode will be supplied by CLS and installed at CLS. To provide access for RF drive, pumping, diagnostics and controls several ports are required in the RF structure. These are shown in the 3D drawing described in section 13. Each port is supplied with a flange for connecting to the external devices. All external devices will be supplied by the CLS.

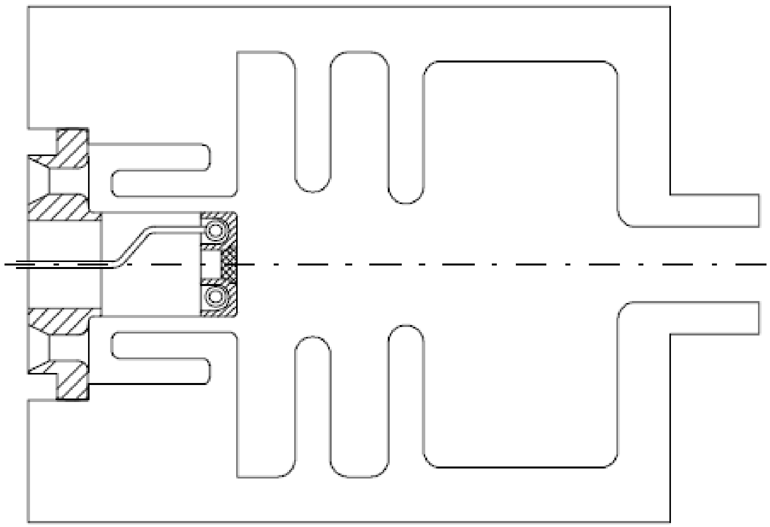


Figure 1: Layout of gun, RF-choke and cathode.

Gun performance specifications are given in table 1. These specifications are based on a MAX IV Laboratory report[2] and may change slightly with the modification to 2856.00 MHz operation. (See also reference 1 and 3.) The electron energy (3.0 MeV in table 1) is sufficiently relativistic (β = 0.985 for 3.0 MeV) to allow injection directly in a standard section of the CLS linac. (I.e., the energy is sufficiently high that the CLS pre-acceleration section 0 is not required.) The MAX IV Laboratory gun operates at 2998.5 MHz. CLS requires a gun that operates at 2856.00 MHz. MAX IV Laboratory shall re-design the gun to operate at the lower frequency. The energy spread from the gun is possibly small enough that no energy filtering is required. If energy filtering is required CLS will be responsible for that system. The beam emittance and bunch length are sufficiently small for efficient injection into the CLS linac.

Table 1. Expected Performance of the 2856 MHz RF gun

Power dissipation	6.0			MW
Q	12500			
Maximum electric field on axis	100			MV/m
Maximum electric field on boundary	120			MV/m
Frequency	2856.00			MHz
Linac operating temperature	45.0 ± 0.1			°C
Minimum beam kinetic energy	3.0 – 3.5			MeV
Rise time	0.42			μs
Input coupling	2			
Coupling between cavities	2.6			
Current from cathode	0	100	600	mA
Beam power	0	0.23	1.38	MW (@2.3 MeV)
Energy spread (nucleus of bunch)	0.13	1.1	18	keV (RMS)
Phase spread (nucleus of bunch)		2		Degrees
Electron pulse length (flat part)	-	200	-	ns (RMS)
Emittance	0.03	0.85	2.0	mm-mrad (RMS) (@ 2.3 MeV)
Emittance, normalized	0.16	4.6	11	mm-mrad (RMS) (@ 2.3 MeV)

6.0 SAFETY PERFORMANCE

CLS will ensure that the RF gun meets all required safety standards.

7.0 WORKPLACE SAFETY AND ENVIRONMENT

CLS will ensure that the RF gun meets all required workplace safety and environmental standards.

8.0 APPLICABLE CODES, STANDARDS AND PROCEDURES

CLS will ensure that the RF gun meets all other CLS specifications.

9.0 DESIGN REVIEW, INSPECTION TESTING AND COMMISSIONING

The modified RF gun design shall be reviewed by CLS staff prior to the commencement of fabrication.

After fabrication:

MAX IV Laboratory will ensure that the RF meets the vacuum requirements for proper operation of the gun. Vacuum test may be performed with a used cathode or suitable blank in the cathode position. Gun will be shipped to CLS for installation of the cathode and testing with a 2856 MHz RF source. Final vacuum testing shall be at CLS after the installation of the cathode. RF gun performance shall meet specifications that are agreed upon during the design review and that may not necessarily be in agreement with table 1. Commissioning of the RF as a source for the CLS linac shall be the responsibility of the CLS.

MAX IV Laboratory shall supply available documentation or verbal instruction for the proper commissioning of the RF gun.

10.0 CALIBRATION

Calibration will occur at the CLS test stand.

11.0 RELIABILITY AND MAINTAINABILITY

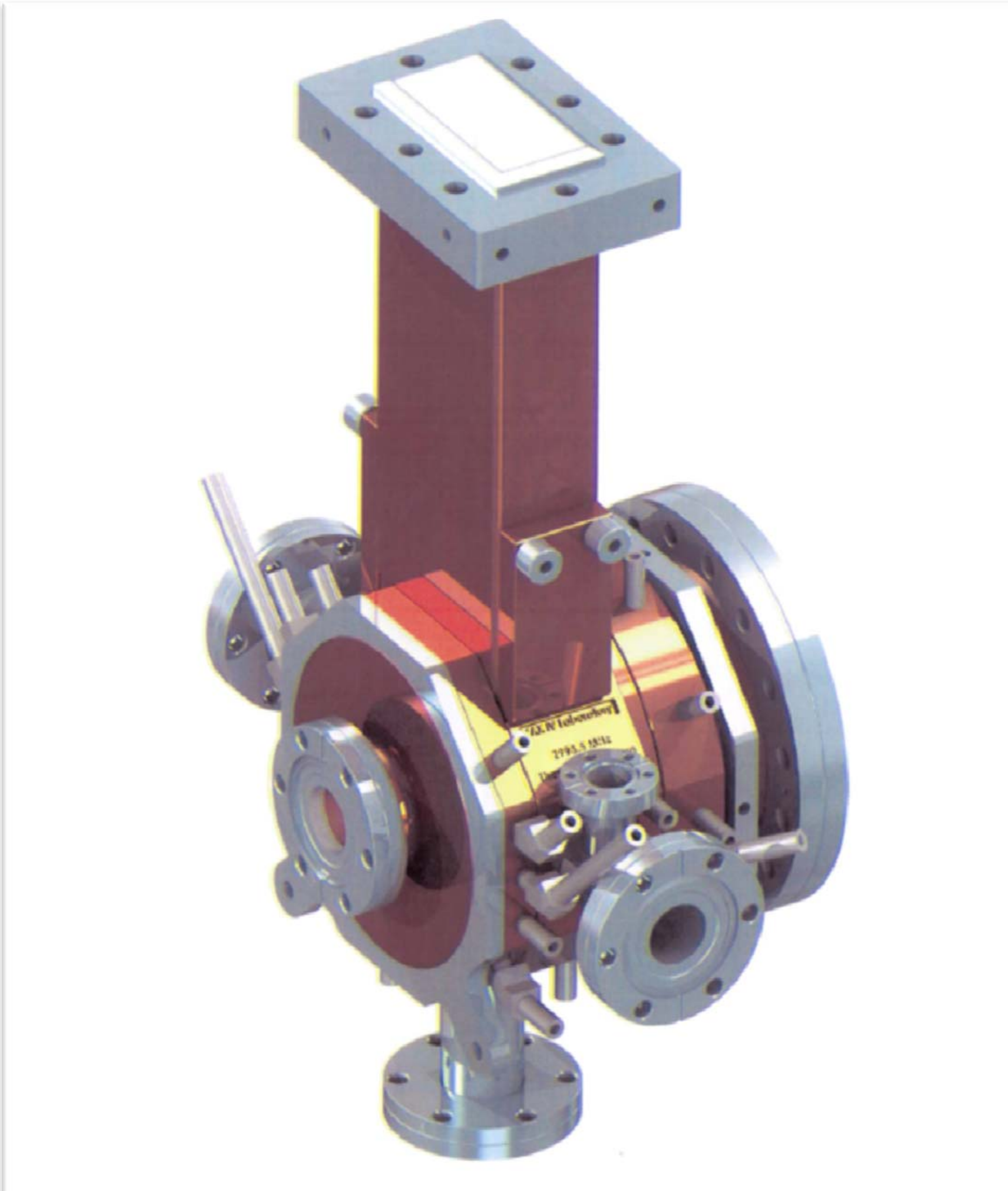
MAX IV Laboratory shall supply available documentation or verbal instructions for ongoing maintenance of the RF gun to ensure continued RF gun reliability.

12.0 OTHER REQUIRMENTS AND CONSTRAINTS

CLS will supply all auxiliary equipment to support operation of the RF gun. This includes ion pump(s), diagnostics, and possible bend magnet system for defining the energy spread from the gun.

13.0 DRAWINGS

An 3D sketch of a 3000 MHz RF gun is shown below. This drawing shows the extent of the RF gun that is to be supplied by MAXIV Laboratory. As part of the 2856 MHz design MAX IV Laboratory will produce drawings suitable for the fabrication of the new gun.



14.0 DOCUMENTATION

The 2856 MHz design shall be documented to report the expected performance of the RF gun.

15.0 REFERENCES

1) B. Anderberg et al, "A NEW 3 GHz RF-GUN STRUCTURE FOR MAX-LABORATORY", EPAC 2000, p. 1684.
2) A. Andersen et al, "An RF-Gun-Driven Recirculated Linac Injector and FEL Driver", Nuclear Instruments and Methods in Physics Research Section A, 445 (2000) p. 413.
3) Sverker Werin, "Thermionic Injector", MAXlab note, 2012.

REVISION HISTORY

Revision	*Date*	*Description*	*Author*
A	2012-09-02	Original Draft	Les Dallin
0	2012-12-12	Issued for use.	Les Dallin